계량경제학 강의 8

다중회귀모형(1)

1. 다중회귀모형

- 우리가 관심을 갖는 많은 문제는 다양한 요인에 의해 영향을 받는다.

예1) 키크는 약

예2) 흡연과 폐암

예3) 교육과 연봉

- 다양한 요인을 모형안에 포함시킨다는 것은 다른 변수들을 통제한다는 것을 의미한다.

1. 다중회귀모형

- 모집단 회귀방정식:

$X_{1i}=1$

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \cdots + \beta_k X_{ki} + e_i$$

$$\boxed{\frac{\partial Y_i}{\partial X_{ji}} = \beta_j} \rightarrow$$ partial derivative. (다른 X control)

- 고전적 가정

① $E(e_i) = 0\,, \forall i$

② $Var(e_i) = E(e_i^2) = \sigma^2\,, \forall i$

③ $Cov(e_i, e_j) = E(e_i e_j) = 0\,, \forall i \neq j$

$e_i \sim \text{iid } N(0, \sigma^2)$

④ $X\!: non - stochastic$

⑤ $No\ exact\ multicollinearity$

(No linear relationship among Xs)

1. 다중회귀모형

⑤ $No\ exact\ multicollinearity$: 다중공선성이 없어야 한다.
즉, 설명변수간의 선형관계가 존재하면 안된다는 것을 의미한다.

예) $X_{3i} = 2X_{4i}$

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + e_i$$

$\parallel 2X_{4i}$

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 (2X_{4i}) + \beta_4 X_{4i} + e_i$$

$$Y_i = \beta_1 \cdot 1 + \beta_2 X_{2i} + (2\beta_3 + \beta_4) X_{4i} + e_i$$

$\Rightarrow \hat{\beta}_1 , \hat{\beta}_2 , \widehat{2\beta_3 + \beta_4}$ 추정가능

$\beta_3 . \beta_4$ 구별할 수 없음

예1) $X_{1i} \sim X_{2i}$

$cov(X_{1i}, X_{2i}) = 0.9 \uparrow$

$\hat{\beta}_1 , \hat{\beta}_2$

$\Rightarrow var(\hat{\beta}_1) \quad var(\hat{\beta}_2)$

$H_0 : \beta_1 = 0$

$t\text{-value} = \dfrac{\hat{\beta}_1 - 0}{\sqrt{var(\hat{\beta}_1)}}$ BiG $\Rightarrow t \downarrow$

2. 다중회귀모형 : OLS 추정량

sample regression equation

- OLS 추정량

$$\min \sum_{i=1}^{n} \hat{e}_i^2 = \sum_{i=1}^{n}(Y_i - \widehat{\beta}_1 - \widehat{\beta}_2 X_{2i} - \cdots - \widehat{\beta}_k X_{ki})^2$$

(residual sum of squares)

1계조건(F.O.C) :

$① \quad \frac{\partial \sum \hat{e}_i^2}{\partial \hat{\beta}_1} = 2\sum(Y_i - \hat{\beta}_1 - \cdots - \hat{\beta}_k X_{ki})(-1) = 0$ $\quad \sum \hat{e}_i$

$② \quad \frac{\partial \sum \hat{e}_i^2}{\partial \hat{\beta}_2} = 2\sum(\;"\;)(-X_{2i}) = 0$ $\quad \sum \hat{e}_i$

$\vdots$

$ⓚ \quad \frac{\partial \sum \hat{e}_i^2}{\partial \hat{\beta}_k} = 2\sum(\;"\;)(-X_{ki}) = 0$ $\quad \sum \hat{e}_i$

$\sum \hat{e}_i = 0$

$\sum \hat{e}_i X_{2i} = 0$

$\vdots$

$\sum \hat{e}_i X_{ki} = 0$

$j = 1, 2, \ldots k$

$\sum \hat{e}_i X_{ji} = 0$

$\hat{e}_i$이 X에 대한 정보를 가지지 않는다.

2. 다중회귀모형 : OLS 추정량

<모집단>

- 행렬의 도입

$$Y_1 = \widehat{\beta_1} + \widehat{\beta_2}X_{21} + \cdots + \widehat{\beta_k}X_{k1} + \widehat{e_1}$$

$$Y_2 = \widehat{\beta_1} + \widehat{\beta_2}X_{22} + \cdots + \widehat{\beta_k}X_{k2} + \widehat{e_2}$$

$$\vdots$$

$$Y_n = \widehat{\beta_1} + \widehat{\beta_2}X_{2n} + \cdots + \widehat{\beta_k}X_{kn} + \widehat{e_n}$$

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & X_{21} & \cdots & X_{k1} \\ 1 & X_{22} & & X_{k2} \\ \vdots & & \ddots & \vdots \\ 1 & X_{2n} & \cdots & X_{kn} \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}$$

Y $n \times 1$ $\quad X$ $n \times k$ $\quad \beta$ $k \times 1$ $\quad e$ $n \times 1$

Y $n \times 1$ $\quad X \quad \beta \quad e$

$$Y = X\beta + e \xrightarrow{\text{sample}} Y = X\hat{\beta} + \hat{e}$$

2. 다중회귀모형 : OLS 추정량

- 행렬의 도입

① $E(e_i) = 0 , \forall i$

② $Var(e_i) = E(e_i^2) = \sigma^2 , \forall i$

③ $Cov(e_i, e_j) = E(e_i e_j) = 0 , \forall i \neq j$

④ $X: non - stochastic$

⑤ $No\ exact\ multicollinearity$

$\Rightarrow$ Column Rank(X) = k

선형 독립인 행이나 열의 최대수

e.g.) $a_1 + a_2 + a_3 = 1$
$2a_1 + a_2 + 3a_3 = 1$
$2a_1 + 2a_2 + 2a_3 = 2$

non-singular matrix

$B \cdot a = h$, $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$

$a = B^{-1}h$

① $\underset{n \times 1}{E(e)} = 0$, $E\begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$

② ii) + iii)

$\underset{n \times n}{Var(e)} \underset{\text{by definition}}{=} E[\underset{n \times 1}{(e - E(e))}\underset{1 \times n}{(e - E(e))'}]$

$= E[ee']$

$= E\left(\begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} [e_1 \cdots e_n]\right)$

$= E\begin{bmatrix} e_1^2 & e_1 e_2 & \cdots & e_1 e_n \\ e_2 e_1 & e_2^2 & & \\ \vdots & & \ddots & \\ e_n e_1 & & & e_n^2 \end{bmatrix}$

$cov(e_i, e_j) = 0 \ \forall i \neq j$

$E(e_i^2) = \sigma^2 \ \forall i$

$\begin{bmatrix} \sigma^2 & & 0 \\ & \sigma^2 & \\ 0 & & \ddots \ \ \sigma^2 \end{bmatrix}$

2. 다중회귀모형 : OLS 추정량

모집단 특징

① $\hat{\beta}, \hat{\sigma}^2$ 추정값 유도 (OLS)

② If Repeated sampling,

$E(\hat{\beta}) = \beta$

$E(\hat{\sigma}^2) = \sigma^2$

③ $Var(\hat{\beta})$

$Var(\hat{\sigma}^2)$

④ $\hat{\beta}, \hat{\sigma}^2$ 분포?

⑤ 가설검정

- OLS 추정량

$$\sum \hat{e}_i^2 = \underset{1\times n}{\hat{e}'}\ \underset{n\times 1}{\hat{e}} \quad (1\times 1)$$

$$\min \sum_{i=1}^{n} \hat{e}_i^2 = \hat{e}'\hat{e} \quad \text{w. r. to } \hat{\beta}$$

$z'z \Rightarrow$ 미분 $2 \cdot z$

1계 조건 :

$$\frac{\partial \hat{e}'\hat{e}}{\partial \hat{\beta}} = \frac{\partial (Y - X\beta)'(Y - X\beta)}{\partial \hat{\beta}} \Rightarrow 2(-X)'(Y - X\hat{\beta}) = 0$$

$$X'(Y - X\hat{\beta}) = 0$$

$$X'Y - X'X\hat{\beta} = 0$$

$$(X'X)\hat{\beta} = X'Y$$

$$\Rightarrow \therefore \hat{\beta} = (X'X)^{-1}X'Y$$

단 $(X'X)^{-1}$: 존재.

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_k \end{bmatrix}$$

$\boxed{\text{Review}}$ pop-reg. equation

$$\underset{n\times1}{y} = \underset{n\times k}{X}\,\underset{k\times1}{\beta} + \underset{n\times1}{e}$$

<가정>

- $e \sim N(0, \sigma^2 I_n)$
- X : non-stochastic
- column Rank$(X) = k$

$$\begin{bmatrix}\sigma^2 & & 0\\ & \ddots & \\ 0 & & \sigma^2\end{bmatrix} = \sigma^2\begin{bmatrix}1 & & 0\\ & \ddots & \\ 0 & & 1\end{bmatrix} = \sigma^2 I_n$$

$$\hat\beta = \begin{bmatrix}\hat\beta_1\\ \vdots \\ \hat\beta_k\end{bmatrix} \qquad \hat\sigma^2 \Rightarrow$$

"by OLS"

$$\Leftrightarrow\; y_i = \beta_1 + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + e_i$$

FOC

$$\frac{\partial\, \hat e'\hat e}{\partial \hat\beta} = 0 \;\Leftrightarrow\; \frac{\partial \sum \hat e_i^2}{\partial \hat\beta_1} = 0,\ \ \vdots\ \ , \frac{\partial \sum \hat e_i^2}{\partial \hat\beta_k} = 0$$

$$\hat e'\hat e = \sum_{i=1}^{n} \hat e_i^2$$

$$\hat\beta = (X'X)^{-1}X'Y$$

3. 다중회귀모형 : OLS 추정량의 분포

$\hat{\beta} = (X'X)^{-1}X'Y$, $Y = X\beta + e$

- $E(\hat{\beta}) = E[\cancel{(X'X)^{-1}X'X}\beta + (X'X)^{-1}X'e]$

$= \beta + E((X'X)^{-1}X'e) = \beta + (X'X)^{-1}X'E(e)$, $E(e) = 0$

$\Rightarrow \hat{\beta} = \beta + (X'X)^{-1}X'e$

$\boxed{\hat{\beta} - \beta = (X'X)^{-1}X'e}$

- $Cov(\hat{\beta}) = E\left((\hat{\beta} - \beta)(\hat{\beta} - \beta)'\right)$

$\beta = E(\hat{\beta})$

$(X'X)^{-1}$: symmetric, $(X'X)^{-1\prime} = (X'X)^{-1}$

$= E[(X'X)^{-1}X'ee'X(X'X)^{-1}]$

$= (X'X)^{-1}X'E(ee')\cancel{X(X'X)^{-1}}$, $E(ee') = \sigma^2 I_n$ (by 가정)

$Cov(\hat{\beta}) = \sigma^2(X'X)^{-1}$

- $e \sim N(0, \sigma^2 I)$라면,

$\hat{\beta}$, $\hat{\sigma}^2 = \dfrac{\hat{e}'\hat{e}}{n-k}$, $Var(\hat{\beta}) = \sigma^2(X'X)^{-1}$

$= \sigma^2 \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ \vdots & \ddots & & \\ a_{k1} & \cdots & & a_{kk} \end{bmatrix}$

$var(\hat{\beta}_j) = \sigma^2 a_{jj}$, $\forall j = 1, \ldots, k$

$cov(\hat{\beta}_j, \hat{\beta}_h) = \sigma^2 a_{jh}$, $\forall j \neq h$

3. 다중회귀모형 : 표본분산

- $\widehat{\sigma^2} = \dfrac{\hat{e}'\hat{e}}{\Box}$

Q) $\Box = n-k$, 왜 $n-k$ 인가?

✓ $E(\hat{\sigma}^2) = \sigma^2$

✓ $E\left(\dfrac{\hat{e}'\hat{e}}{\Box}\right) = \sigma^2$ 목표!

✓ $E(\hat{e}'\hat{e}) = \Box\,\sigma^2$

$\hat{e} = Y - X\hat{\beta}$

$= Y - X(X'X)^{-1}X'Y$

$= \underbrace{(I_n - X(X'X)^{-1}X')}_{M_x} \underbrace{Y}_{X\beta + e}$

$[I_n - X(X'X)^{-1}X'] \cdot X = X - X = \Phi$

$= (I_n - X(X'X)^{-1}X')e$

$\Rightarrow \hat{e} = M_x \cdot e$

$\boxed{M_x : \text{symmetric, idempotent}}$

Note

- $M_x = I_n - X(X'X)^{-1}X'$

① $M_x' = M_x$: symmetric

② $M_x \cdot M_x = M_x$: idempotent

- $M_x' M_x = M_x$

※ Note: Trace(A)

trace of a square matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} \\ \vdots & & & \vdots \\ a_{j1} & \cdots & & a_{jj} \end{bmatrix}$$

$tr(A) = \sum_{j=1}^{k} a_{jj}$

- $tr(\underset{1\times1}{k}) = k$
- $tr(k \cdot A) = k \cdot tr(A)$
- $tr(A+B) = tr(A) + tr(B)$

$tr(AB) =$

$\hat{e}'\hat{e} = (M_x e)'(M_x e)$

$= (e' M_x')(M_x e)$, $M_x' M_x = M_x$

$\hat{e}'\hat{e} = e' M_x e$

$E(\underset{1\times n}{\hat{e}'}\underset{n\times1}{\hat{e}}) = E(e' M_x e)$

$= E(tr(e' M_x e))$

$= E(tr(M_x e e'))$

$= \sigma^2 \{tr(I_n - X(X'X)^{-1}X')\}$ ($\because X$: non-stochastic)

$= \sigma^2\, tr(I_n) - tr(X(X'X)^{-1}X')$ ($\because Rank(X) = k$)

$= \sigma^2 (n-k)$

$\therefore E(\hat{e}'\hat{e}) = \sigma^2(n-k)$

$E\left(\dfrac{\hat{e}'\hat{e}}{n-k}\right) = \sigma^2$, $\dfrac{\hat{e}'\hat{e}}{n-k} = \hat{\sigma}^2$

$\hat{\sigma}^2 = \dfrac{\hat{e}'\hat{e}}{n-k}$ 은 σ^2의 unbiased estimator,

Summary

- 상수항만 있는 경우

- 단순회귀분석

- 다중회귀분석